

Interview Questions — Introduction to OOPS Programming

Total Questions: 40

1. What is the main difference between procedural programming and object-oriented programming (OOP)?

Hint: Think about how data and functions are treated in each approach.

2. Why is global data in procedural languages like C considered a limitation?

Hint: Consider what happens to data security and integrity when any part of the program can access or modify it.

3. Explain how encapsulation in OOP helps solve problems found in procedural programming.
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4. Imagine you are building a simple task list application. How would the approach differ if you wrote it in C versus using OOP concepts?
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5. What is a class in OOP, and how does it differ from an object?

Hint: One is a blueprint, the other is a real instance.

6. Write the basic syntax to define a class called Note with members: title, date, and isDone.
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7. How would you create an object of the Note class and set its title to 'Interview Prep'?
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8. What are member functions in a class, and why are they useful?
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9. What is a constructor in OOP, and when is it called?

Hint: Think about what happens when you create a new object.

10. Explain the difference between a default constructor and a parameterized constructor.
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11. Write a parameterized constructor for a class Content that sets the title and platform.
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12. What is a copy constructor, and why might you need one?
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13. When is a destructor called in the lifecycle of an object?

Hint: Consider what happens when an object goes out of scope or is deleted.

14. Give an example of why you might use a destructor in a program that saves user data automatically upon exit.

15. What is inheritance in OOP, and how does it help with code reuse?

16. Explain the difference between single inheritance and multilevel inheritance with examples.

17. How would you represent a ContentCreator class that has YouTuber and Podcaster as specialized classes using inheritance?

18. What is hierarchical inheritance, and can you give a scenario where it might be useful?

19. What is polymorphism in OOP?

Hint: Think about how the same function name can behave differently.

20. Explain the difference between compile-time (function overloading) and runtime (function overriding) polymorphism.

21. Suppose you have a method uploadContent() that behaves differently for YouTube and Instagram. How would you achieve this using polymorphism?

22. What are access modifiers in OOP, and why are they important?

Hint: Consider how you control which parts of your code can access certain data.

23. Explain the difference between public, private, and protected access modifiers.

24. What is encapsulation, and how do access modifiers help achieve it?

25. Define abstraction in OOP and explain the purpose of abstract classes.

Hint: Focus on hiding implementation details and showing only the necessary features.

26. What is a pure virtual function? Give an example.

27. How would you design a Platform class with an abstract upload() method that must be implemented by all derived classes?

28. What is file handling in C++ and why is it important in applications?

Hint: Think about how programs store data even after they close.

29. Which classes are used in C++ for reading from and writing to files?

Hint: Focus on stream classes.

30. Write a code snippet to save the title and views of a Content object to a text file.

31. How would you read analytics data from a file and display it on the console?

32. Imagine you are building a mini project: 'Creator Dashboard Lite'. What classes and features would you include to manage content items?

Hint: Think about what data a content item needs and what actions a user might perform.

33. Describe how you would implement the ability to add, edit, and delete content items in your Creator Dashboard Lite project.

34. Explain how file handling would be integrated into your Creator Dashboard Lite app to store and retrieve user data.

35. Suppose your Creator Dashboard Lite app crashes before saving changes. How could you modify your design to minimize data loss?

Hint: Consider when and how often you write data to the file.

36. How could you use an AI tool like ChatGPT or GitHub Copilot to help you write or debug OOP code for your mini project?

Hint: Think about generating code snippets, fixing errors, or understanding concepts.

37. If an AI assistant suggests a piece of code for file handling, what steps would you take to ensure it is correct and safe to use in your application?

38. Integrative: How does encapsulation support file handling when saving and loading private data members of a class?

39. Integrative: In your Creator Dashboard Lite, how would you use inheritance and polymorphism to support multiple types of content platforms (like YouTube, Instagram, Podcast) with different upload behaviors?

- 40.** Integrative: Describe a scenario where using both constructors (for initialization) and file handling (for persistent data) are necessary in the same application.
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